ENGINE CONTROL UNIT · KIRLOSKAR 3GK550ETA 4SR1

Genset ECU Field Handbook

Everything known about replacing the engine control unit on a 25 kVA diesel genset — the machine, the 94-pin interface, every circuit with its simulated result, and the two risks that decide whether this ever runs.

Rev 1 13 Sep 2026 17 circuits verified 10 open unknowns Source: 08-GP3-60-001

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01 The machine

The engine is a **Kirloskar 3GK550ETA 4SR1** — three cylinders, turbocharged and aftercooled, common-rail injection, cooled EGR with a diesel oxidation catalyst and no SCR. **26.5 kW / 36 hp at 1500 rpm**, governing at constant speed to make 50 Hz. Application code `GK3.8703`, built 12 September 2025.

That constant speed matters more than it first appears, and it comes up twice in this handbook: once because it makes the timing problem easier than an automotive engine's, and once because the genset's own output frequency is rigidly tied to shaft speed, which turns out to be what is left of the runaway protection.

The panel controller is a **Deep Sea Electronics DSE4522 MKII AMF**, India special — *not* the Kirloskar KG640C the OEM wiring diagram shows. That distinction is not academic; §04 explains what it cost.

PARAMETER	VALUE	STATUS
Engine	`3GK550ETA 4SR1 · app code GK3.8703`	Confirmed — rating plate
Rated output	`26.5 kW / 36 hp @ 1500 rpm · ISO 3046`	Confirmed — rating plate
Type approval	`ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24`	Confirmed — rating plate
Genset	`25 kVA, 415 V 3-phase, 50 Hz, 4-pole alternator`	Confirmed — controller config
Fuel system	`Common rail, metering unit + 3 injectors`	Confirmed
Aftertreatment	`Cooled EGR + DOC — no SCR, no DEF, no DPF`	Confirmed
Emissions tier	`CPCB IV+ (IGES4) · GSR 804(E), 3 Nov 2022`	Confirmed
Electrical	`12 V, 75 Ah`	Confirmed
Panel controller	`DSE4522 MKII AMF (India SP) · 4522-001-01`	Confirmed — photographed
ECU connector	`Bosch 1 928 405 192 / 194 · code C · 94-way`	Confirmed — markings
Our MCU	`NXP FS32K148HAT0MLQT, 144-LQFP`	Selected

ON THE NAME "GK550" — WE GOT THIS WRONG

The engine was first described to us as a **GK550**. An earlier revision of this handbook said no Kirloskar document used that string, that it was a garbled run-together of the genset prefix and the engine family number, and that it "will find nothing" when ordering parts. The rating plate reads `3GK550ETA 4SR1`. **The original description was correct and our inference was wrong.** The reasoning that produced it was sound — five hardware constraints read off the wiring diagram, all of which match the real engine — but it converged on a different Kirloskar engine that happens to satisfy the same five. Use `3GK550ETA 4SR1` and `GK3.8703`.

02 The 94-pin interface

The OEM ECU presents one 94-pin connector. The troubleshooting diagram traces **44 of those pins**; the other 50 — power grounds, unused positions, whatever drives the intake throttle — are simply not on a document written for field technicians. A genuine drop-in replacement needs the connector part number to get the rest.

Every wire in the diagram is colour-coded by signal class, and that code is load-bearing rather than cosmetic: it is how we established that the discrete inputs are switched-battery rather than switched-ground, and that the fuel metering unit is driven low-side. The same colours are used below.

- DC positive · sensor supply · injector high side
- DC negative · sensor ground · injector low side
- Analog in · PWM out
- Digital in · frequency
- CAN H
- CAN L

PIN	SIGNAL	CLASS	FRONT-END
21	**Battery + feed**	supply	EMI filter → fuse → TVS → reverse-battery FET
04 / 06	**V_V_BAT_1R / 2R**	supply	120 k / 10 k divider → ADC · 10 A fused each · 04 feeds the injector, 06 the EGR bridge (POWER-PATH, 8 Oct 2026)
09	**Synchronization ground**	?	Function unresolved
11 / 41 / 79 / 34	**Boost pressure + air temp**	analog	Combo sensor, 4-pin, shared supply and ground
33	**Coolant temperature**	analog	NTC, pull-up divider · ground splices to 34
32 / 35 / 08	**Rail pressure**	analog	Ratiometric · dedicated sensor ground
39 / 80	**Oil pressure**	analog	Ratiometric · ground splices to 36

PIN	SIGNAL	CLASS	FRONT-END
15 / 37 / 36	**EGR position**	analog	Potentiometer, shares EGR connector
52 / 74 / 30	**Crankshaft speed**	frequency	Variable reluctance, differential, twisted pair
45 / 46 / 44	**Camshaft position**	frequency	Hall effect, already digital
43 / 23	**Coolant switch · droop switch**	digital	Switch-to-return + debounce
20 / 24 / 71	**Audio abort · override · ignition**	switched batt	**Active high** — zener clamp, see §03
01 / 02	**G_G_B AT1 / AT2**	ground	Switch returns, or main power grounds — unresolved
88	**Fuel metering unit**	PWM low side	Low-side driver + current sense
59 / 81	**EGR high / low**	PWM	H-bridge, position feedback on pin 37
03	**Injector high side — bank A**	high side	Cylinders **1 and 3**, joined at splice V5
05	**Injector high side — bank B**	high side	Cylinder 2
73 / 07 / 29	**Injector low side — cyl 1 / 3 / 2**	low side	Current-controlled peak-and-hold
50 / 69	**Main relay · buzzer relay**	low side	N-FET + flyback diode
55 / 77	**CAN channel 1**	CAN H CAN L	To the KG640C
87 / 86	**CAN channel 2**	CAN H CAN L	Also wired in the OEM system

> **INJECTORS SHARE THEIR HIGH SIDES**
>
> Pin 03 feeds cylinders 1 *and* 3. Only the low sides are individual. That is five pins, not six, and two boost-fed bank switches rather than three — which matters, because the boost stage is the expensive part of the board. It is safe because on a three-cylinder four-stroke any two cylinders are 240 crank degrees apart, or 26.7 ms at 1500 rpm, against an injection lasting perhaps 3 ms. That separation holds under either candidate firing order, so the unknown firing order stops mattering.

Signals that are not on this connector

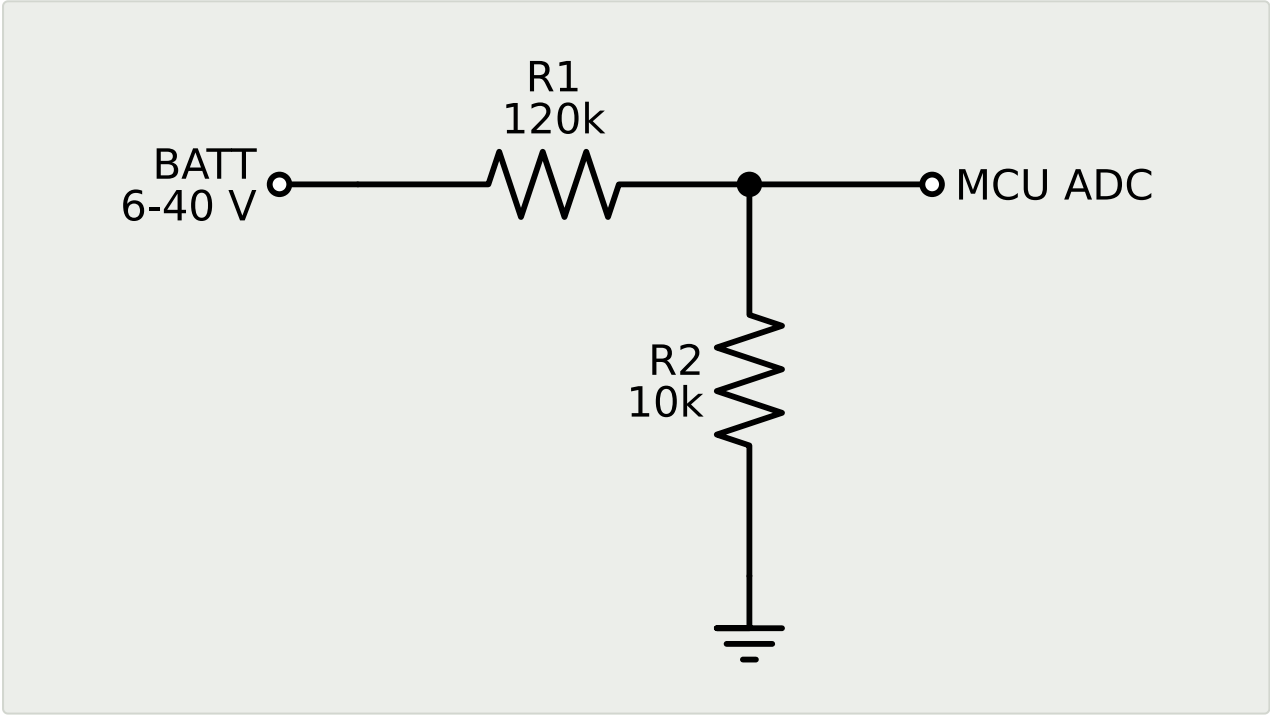
The fuel level sender and the water-in-fuel switch both appear in the wiring diagram, and neither reaches the ECU — they route through the XC11/XC1 harness to the panel. They belong to the GCU. Earlier design passes had built front-ends for both.

Also absent from the entire system: any magnetic pickup, any DEF or urea sensor, any LIN bus, any SCR, and any separate overspeed switch. What the diagram labels `OVERSIDE SWITCH` on pin 24 traces to `-S7 Override SS SW` — one component, not two.

03 The circuits

Every circuit below exists as a SPICE netlist that runs, with checks that are claims about the circuit rather than about the simulator. A failing check means the design is wrong. All six currently pass — but two of them only after simulation caught errors that had survived four rounds of document review.

Battery voltage sense pins 04, 06



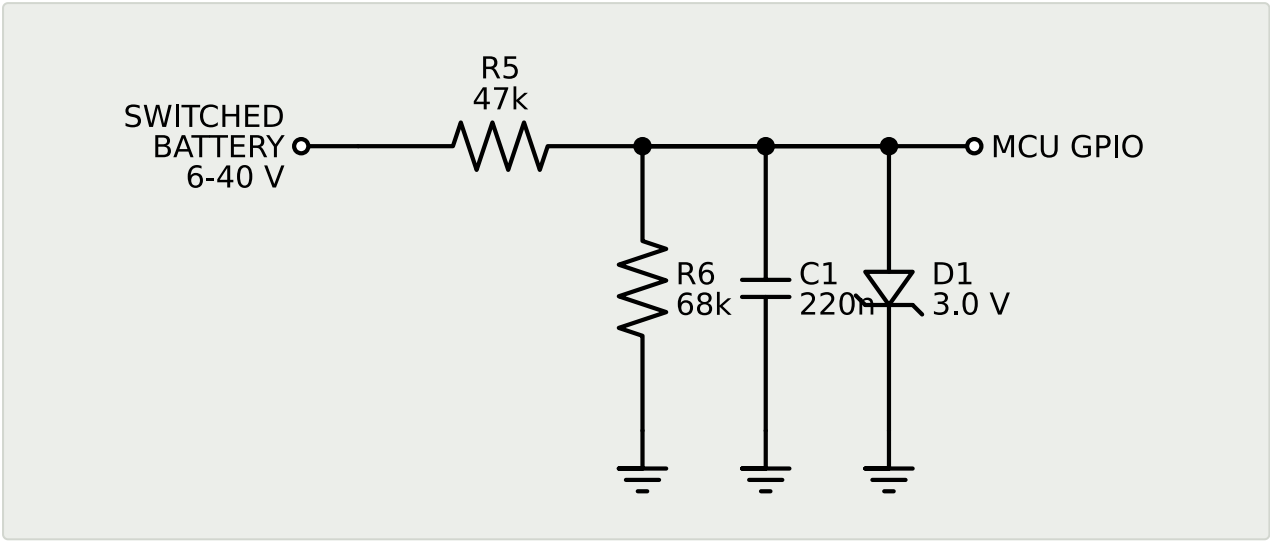
A plain resistive divider scaling the 6–40 V rail into the ADC. Two channels, one per fused rail, which doubles as fuse-integrity monitoring: voltage present downstream means the fuse is intact.

The inherited value was **75 k / 10 k**, and it would have destroyed an ADC pin. That ratio was sized for a 5 V ADC; this MCU's analog supply is 3.3 V. At the top of the input range it delivers 4.69 V into a 3.3 V pin. Corrected to **120 k / 10 k**.

PASS	Peak output at 40 V in	3.069 V
PASS	Headroom below the 3.3 V rail	0.231 V
PASS	Reading at the 6 V cranking dip	0.462 V · 573 LSB
PASS	Linearity residual	< 1 µV

Discrete switch input

pins 20, 24, 71 · active high



These are switched-battery inputs: the switch connects the pin to battery, not to ground. Earlier documents had them as pull-up-to-5 V with the switch closing to ground — the opposite sense, with a front-end that cannot survive battery voltage.

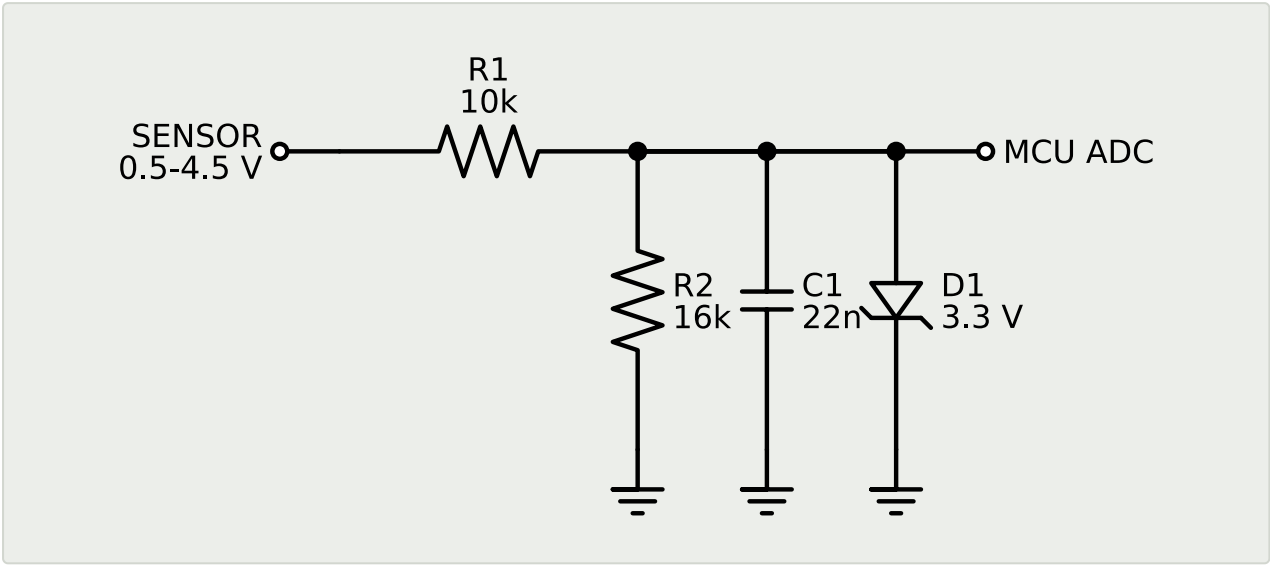
This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 2.31/6 = 0.385. Staying under 3.3 V at 40 V needs it below 3.3/40 = 0.0825. Those do not overlap, so no fixed ratio works at both ends — the original was impossible, not merely mis-valued.

A 3.0 V zener sets the clamp level and the divider only biases into it, so the logic level stops tracking the battery and goes flat.

PASS	Level at 6 V cranking dip	2.863 V
PASS	Level at 40 V load dump	2.954 V
PASS	Spread across the whole range	92 mV
PASS	Threshold crossings during a 5-bounce contact	0
PASS	Debounce time constant	6.11 ms

Debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two button presses.

Ratiometric sensor front-end pins 41, 35, 80, 37



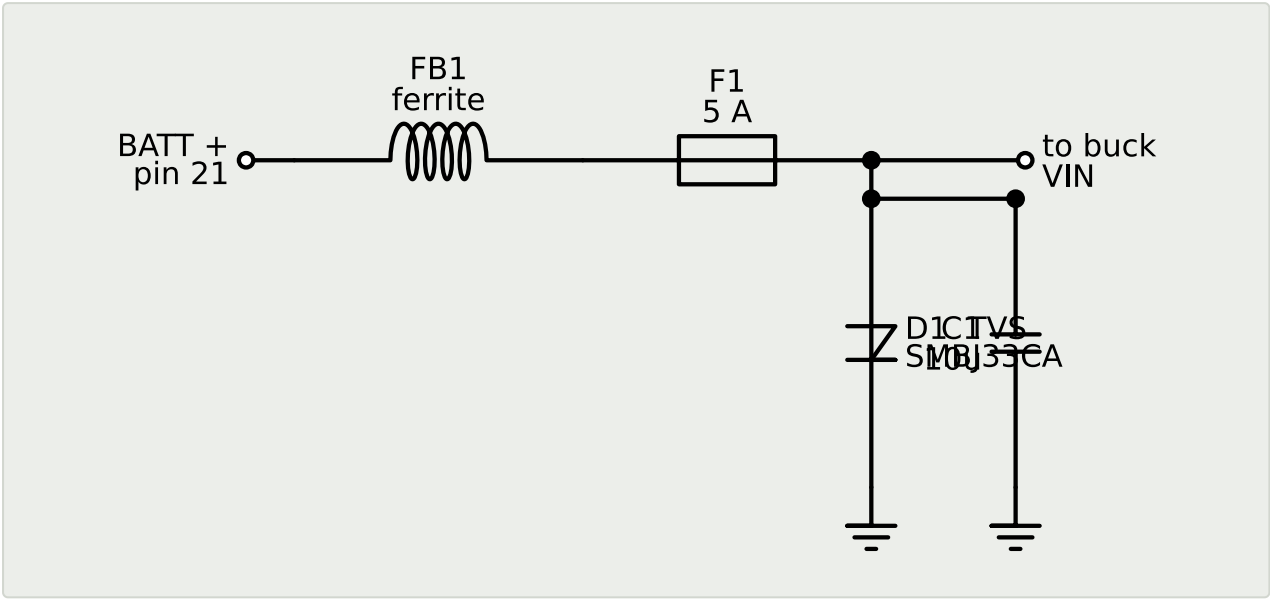
Boost pressure, rail pressure, oil pressure and EGR position are all 0.5–4.5 V ratiometric sensors — their output is a fraction of their own 5 V supply, so the ADC reference has to track that same rail or a drifting supply reads as a pressure change.

Full scale is deliberately placed at 2.77 V, below the 3.3 V clamp rather than hard against the ADC ceiling. That costs about 4% of span and buys a front-end that a 40 V harness short cannot damage.

PASS	Sensor zero (0.5 V) maps to	0.308 V
PASS	Sensor full scale (4.5 V) maps to	2.769 V
PASS	ADC span used	74.6 %
PASS	Low-pass corner	1175 Hz
PASS	Loss at the sensor's own 100 Hz	0.03 dB
PASS	Attenuation at 100 kHz switching noise	38.6 dB
PASS	Pin voltage on a 40 V harness short	3.31 V · 3.7 mA

Transient clamp

battery input · ISO 7637-2 pulse 2a

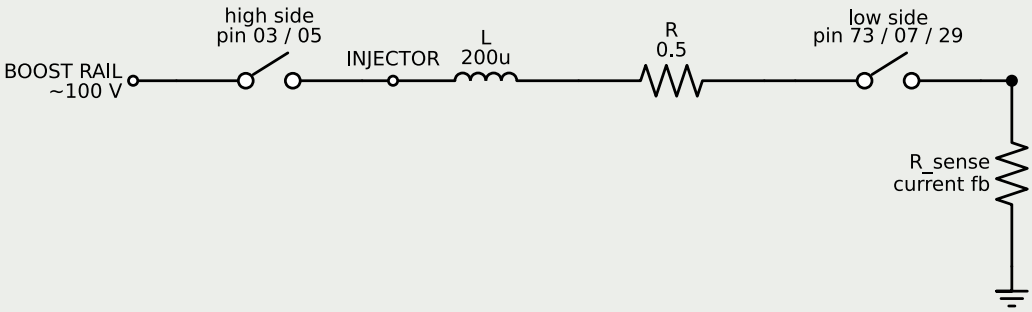


This is the circuit that decides whether the board survives its electrical environment, so it gets simulated rather than assumed. Pulse 2a models a sudden current interruption in another device sharing the harness: **+112 V through 2 Ω for 50 µs**, at Level IV severity.

PASS	Clamped voltage at the buck input	50.0 V
PASS	Margin inside the LM5164's 100 V rating	2.0 ×
PASS	Recovery after the pulse	12.98 V

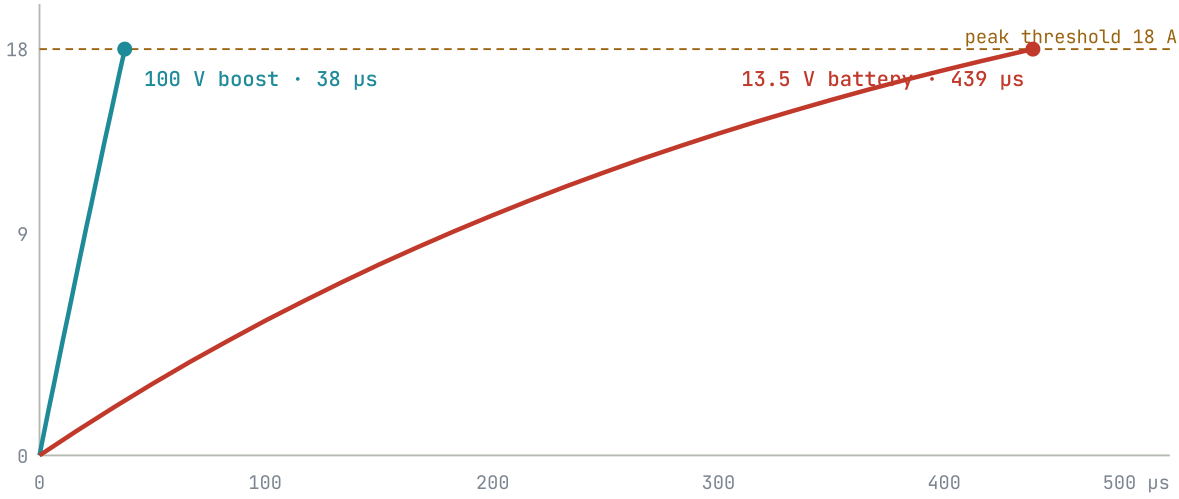
> **A SPEC TARGET THAT SHOULD BE DROPPED**
>
> Earlier documents asked for a clamp "under 42 V". An SMBJ33CA cannot do that — the real part is specified at 53.3 V clamping at rated pulse current, so 50 V here is the component behaving correctly. The 42 V figure was never derivable from a datasheet. Chasing it with a lower-voltage TVS would make the clamp conduct during a legitimate suppressed load dump and force it to absorb that energy. Drop the target; keep the part.

Injector drive — why the boost stage exists pins 03, 05 · 73, 07, 29



An injector is an inductor with resistance, so current through it cannot change instantly — `di/dt = V/L`. The needle does not lift until current reaches its peak threshold, so every microsecond spent getting there is dead time in series with every injection event.

This is the single most expensive stage on the board. Here is its justification as a measurement rather than an assertion.

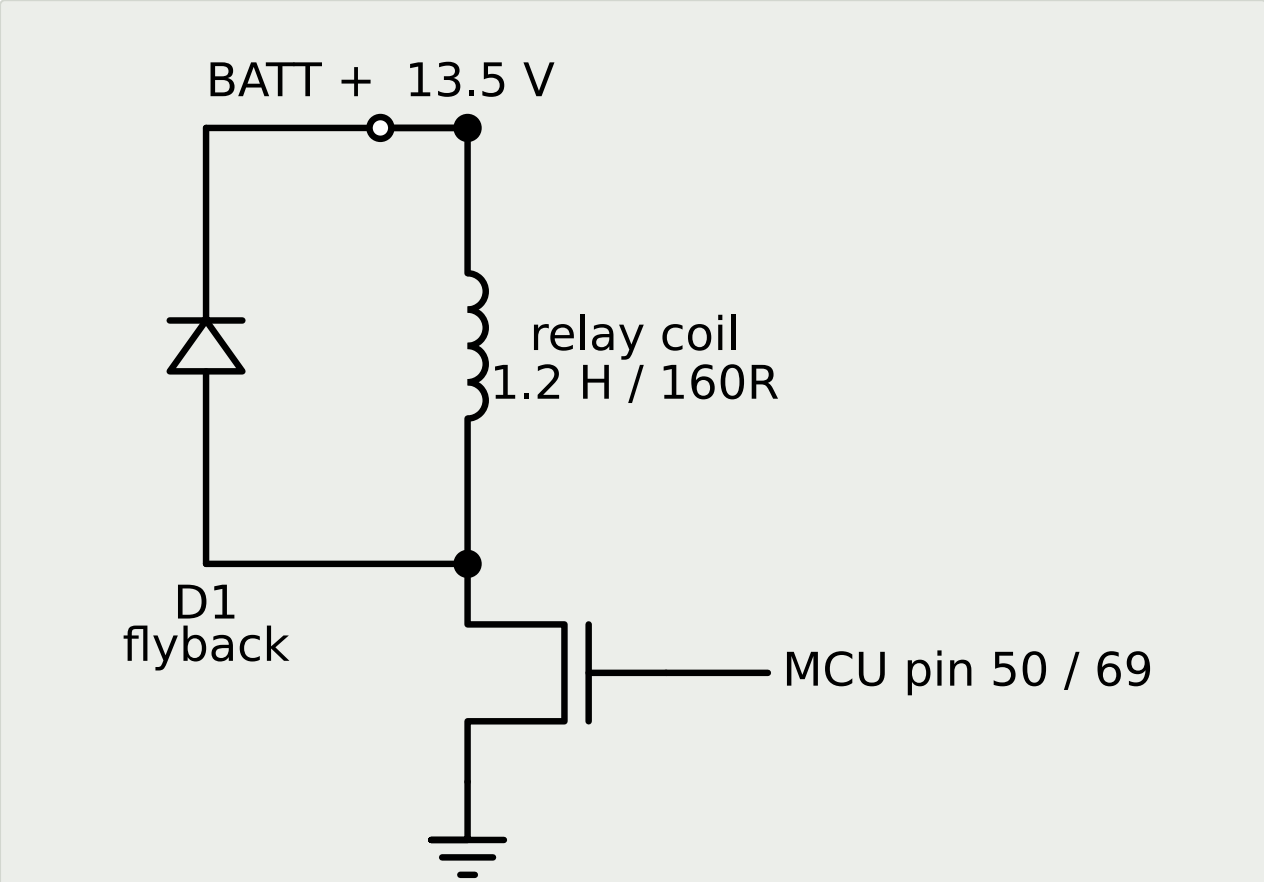


Injector current reaching its 18 A peak threshold. Modelled at 200 µH / 0.5 Ω, the mid-range of the envelope for solenoid common-rail injectors of this class.

PASS	Time to 18 A from a 100 V boost rail	37.8 µs
PASS	Time to 18 A from the 13.5 V battery	439.5 µs
PASS	Boost is faster by	11.6 ×
PASS	Boost drive, as a share of a 1 ms injection	3.8 %
PASS	Battery drive, as a share of a 1 ms injection	44.0 %

Without the boost stage, nearly half of the shortest injection event is spent merely lifting the needle. That is why no production common-rail ECU drives injectors from the battery rail, and why the boost converter is not optional.

Relay driver `pins 50, 69`



A low-side N-FET switching a relay coil. The flyback diode is the entire point: a coil is an inductor, and interrupting its current generates `-L·di/dt` across it. The diode gives the collapsing field somewhere to circulate.

PASS	Drain peak with the flyback diode	14.3 V
PASS	Drain peak without it	unbounded

The simulator reports hundreds of kilovolts for the undiode case. Read that as a model artifact — an ideal switch opening instantly across an inductor implies an unbounded voltage. A real MOSFET avalanches at its breakdown rating and turns the coil's stored energy into heat in the die, once per relay operation, until it fails. The magnitude is fictional; the conclusion is not.

04 Stopping a runaway

A 25 kVA set in runaway is a mechanical hazard to anyone near it. The failure that matters is a controller hanging with fuel still flowing — and a controller cannot be its own protection against that.

This section has now been wrong twice, in opposite directions, and the second time is worth understanding before trusting the third.

We first read the system as sending engine speed to the panel over CAN only, which would leave overspeed protection depending on the very microcontroller and bus most likely to have failed. We then found the panel controller had an independent alternator-frequency speed path, concluded the pessimism was unwarranted, and relaxed the requirement for a standalone trip module. **That second reading was about the wrong device.**

The OEM wiring diagram shows a Kirloskar `KG640C`. The controller actually fitted, photographed front and back, is a **Deep Sea Electronics DSE4522 MKII AMF**. And DSE's own manual for its siblings says plainly that with a CAN engine, *"compatible ECU's receive the start command via CAN and transmit the engine speed to the DSE controller"*.

So engine speed reaches this controller from us. Its configured 1710 rpm overspeed shutdown reads a number our ECU sends, and a hung ECU transmitting a stale, plausible speed defeats it completely. The first, pessimistic reading was closer to the truth than the second.

> WHAT THIS CHANGED
>
> The standalone magnetic-pickup trip module — a GAC SSW675 or a Murphy HD9063, both with Indian distributors — goes back to **mandatory**. The relaxation to "recommended" was bought entirely by the independent-speed claim, and that claim described a controller that is not on this machine.

One independent path does survive, and it is worth having — but it is a different mechanism and a more conditional one. The DSE4522 measures **generator frequency** directly at its terminals 21–24, straight off the alternator's three phases, without touching CAN at all.

Generator frequency	Over-frequency shutdown	Fuel relay, terminal 3

Measured directly at terminals 21–24. On a 4-pole alternator with no slip, 1500 rpm *is* 50 Hz.

→

Configured to trip at `56.0 Hz` — that is **1680 rpm**, below the CAN-dependent 1710 rpm trip.

→

DC Output A. 5 A continuous — a coil driver, consistent with operating the 70 A `-13RB1` contactor.

→

Fuel system stays powered

Measured 18 Sep 2026: the relay signals, it does not remove power.

The chain ends here, and it ends broken.

No link before the last one touches this ECU, so a hung ECU does not defeat them. But note what is being relied on: a *generator* protection doing the work of an *engine* protection. It inherits that mechanism's dependencies — the alternator must be excited and producing measurable output. An excitation failure removes it. That is a narrower guarantee than a magnetic pickup watching the flywheel, and it should not be described as equivalent to one.

THE GATE CLOSED, AND IT CLOSED BADLY

Every protective action this controller can take ends at the same place: it de-energises the fuel relay on terminal 3. The question was whether those contacts remove supply, or merely signal. **Measured at the machine on 18 September 2026: they signal.**

So the chain terminates in the failure it was meant to catch. Read together with the CAN finding above, the two compound into one sentence worth being blunt about: **the controller either cannot see the fault or cannot act on it.** Overspeed it reads from our CAN, so a hung ECU hides it. Over-frequency it measures for itself — and then acts through a relay that only signals.

Nothing on this machine currently interrupts fuelling without the cooperation of the ECU being protected against. That is what the standalone trip module now exists to fix, and it is why its contacts have to sit in series with the fuel metering unit's own battery feed through fuse `8F1` — not in the GCU relay chain, which has just failed this exact test.

One caveat kept for honesty: the contact-type check that disambiguates a normally-closed relay is not confirmed to have been done first. It is still wanted. Nothing above changes if it comes back the other way, because the unsafe reading is already the one being designed to.

AND THE ONE NOBODY CONSIDERED

A diesel that starts burning its own lubricating oil — through a failed turbo seal, or crankcase fumes drawn into the intake — has a fuel supply that **none of the protections above can touch**. Every one of them cuts diesel or electrical power. The standard countermeasure is a mechanical **air-intake shutoff valve**, and no such device appears anywhere in this design.

It was never considered and rejected. It was never considered. Whether to fit one is a decision for the machine's owners — the part is mechanical, not electronic — but it should be a decision rather than an omission.

05 The two real risks

Neither is a hardware problem, and neither is solved by better circuit design. They are the two things that decide whether this project ends with a working board on a bench or a running genset.

You cannot buy the injector calibration

The drive stage can be designed, built, and fully verified against a known inductive load. That is real engineering progress and it should proceed.

What it cannot do is deliver the *right amount of fuel*. Producing an electrically correct current pulse is not the same as knowing how much fuel that pulse injected — and that mapping, fuel mass against pulse width across the rail pressure range, is proprietary flow-bench data tied to a specific injector's hydraulic geometry. Bosch and Denso ship it with their own ECUs and not otherwise.

It gets harder in the **ballistic region**, where pulses are too short for the needle to reach full lift and delivery becomes steeply non-linear — which is exactly where a genset at light load spends its time. And empirical tuning on a running engine is not a fallback: at 19.7:1 compression, wrong fuelling risks combustion damage and turbocharger overspeed long before trial and error converges.

ROUTE TO THE DATA	COST	LEGALITY
Supplier licence	Structured around OEM production volumes. Realistically out of reach at this scale.	The sanctioned route.
Measure it yourself	Buy injectors, measure delivery directly on a rate-of-injection rig — Bosch tube or Zeuch method. Bounded and controllable with access to a fuel-injection lab.	Entirely legal. First-party measurement of purchased hardware.
Extract the OEM calibration	Cheapest in engineering time if a donor ECU exists, but the binary is undocumented and likely protected.	Contested. Implicates the manufacturer's copyright. Not recommended without legal clearance.

THE ONE WORTH PURSUING

Route two. It is legal, the cost is bounded, and a university with a fuel-injection lab is a far more reachable thing than a Bosch development agreement.

The regulatory position

This is an emissions-certified genset in service. CPCB IV+ type approval attaches to an engine family at the manufacturer level — it is not a per-unit certificate that travels with the machine regardless of what is done to it.

READ THIS BEFORE PLANNING DEPLOYMENT

CPCB's own RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy **must not be altered** — and that is under CPCB's own sanctioned retrofit scheme, which separately excludes modifications limited to engine control. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not touch the engineering. Designing the board, building it, and bench-testing it against simulated crank and cam signals are all unaffected. It bears on putting it into service on a genset that is in use.

And it compounds the calibration problem in a way neither risk shows alone: this engine meets its emissions limits *partly through calibration* — EGR scheduling and injection timing — not through hardware alone. A replacement ECU can therefore move it out of compliance without a single component changing.

This is a decision for the project's owners to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and had not previously been examined — not to make the decision for anyone.

06 What we don't know

Nothing on this list may be guessed into copper. Most of it needs somebody at the machine with a multimeter and a camera rather than more research — which is worth knowing, because it means one site visit closes most of it.

Four closed on 18 September, from photographs and one manufacturer's manual. Three of the four closed *against* something this project had inferred, which is a fair measure of how much the desk research was carrying.

#	CLOSED	ANSWER
U11	**Engine identity**	3GK550ETA 4SR1, app code GK3.8703 — rating plate. **Our inferred model number was wrong.**
U13	**Which controller is fitted**	DSE4522 MKII AMF (India SP), serial 11021794 — **not the KG640C** the diagram shows. Cost us §04's argument.
U1	**The connector**	Bosch 1 928 405 192 / 194, **code C** keying. 94 ways: power 1–8, then 9–28, 29–50, 51–72, 73–94.
U3	**Does the controller have a speed pickup?**	No — DSE's terminal list runs 1–35 with no pickup input.

#	STILL OPEN	BLOCKS	HOW IT CLOSES
U12	**Does the fuel relay remove power, or only signal?**	**CLOSED 18 Sep 2026 — it only signals.** §04's safety gate, resolved against us	Closed by measurement. The trip module becomes the only ECU-independent shutdown.
U4	**Exact J1939 PGN set**	CAN firmware	Mostly answered — addresses 234 / 44, and the controller wants oil pressure, coolant temp, coolant level and speed from us. Log the bus for the rest.
U2	**Boost air temp — NTC or ratiometric?**	One sensor front-end	One resistance reading across the sensor, engine cold

#	STILL OPEN	BLOCKS	HOW IT CLOSES
U5	**Injector part number**	Boost rail target, not the driver itself	Read the injector body
U6	**What pin 09 "synchronization ground" does**	Ground plan	Continuity, battery disconnected
U7	**Are pins 04/06 sense-only or load feeds?**	No longer blocking: the board now uses them as load feeds (POWER-PATH, 8 Oct 2026)	Measure current at the fuse
U8	**Is the 6-pin intake throttle on the ECU?**	Whether a throttle driver is in scope	Follow the cable by hand
U9	**Where the catalyst temp sensor goes**	Whether an EGT front-end is needed	Trace on the engine
U10	**Which signal uses SENT**	Possible digital sensor front-end	Inspect the harness
U14	**Pre-heat / post-heat — controller or ECU?**	Whether a heater output is in scope. We have none.	Trace the heater supply. The controller config enables both at 50 °C.
U15	**What "CRS-878" denotes**	Possibly bears on the injector	Ask KOEL, or find DSE's engine-profile list
—	**Are pins 01/02 switch returns or power grounds?**	Ground plan	Continuity to battery negative

U12 NOW STANDS ALONE

With the controller identified, every protective action it can take ends at one relay. U12 asks what that relay's contacts actually carry, and the whole of §04's second line of defence rests on the answer. It is a multimeter and ten minutes, and nothing else on this list substitutes for it.

07 Running the simulations

You already have a SPICE engine. KiCad ships `libngspice`, and the harness in the repository drives it directly, so the circuits run and check themselves without installing anything further.

```
.venv/bin/python sim/run_sim.py              # every block
.venv/bin/python sim/run_sim.py injector     # blocks matching "injector"
```

Output is one line per claim, green or red. Because the claims are about the circuit — "reads logic high at a 6 V cranking dip", "never exceeds the MCU's absolute maximum" — a red line means the design is wrong, not that the simulator is unhappy.

The netlists are plain SPICE and are not tied to this harness. They open directly in **KiCad**'s built-in simulator (Tools → Simulator), in **LTspice**, or in a standalone **ngspice**. Each carries its own `.control` block, so it runs without further setup.

Read the comments inside each netlist before the netlist itself. They carry the reasoning — why the discrete input is not a divider, why the sensor full scale sits below the clamp, why the 42 V clamp target should be abandoned — which the schematic alone cannot tell you.